

23/07/26

SIMATS Online Programming Lab

PYTHON PROGRAMMING

Suba Shree K
192521396RD

```
l=3
o=2
r=1
w=1
```

Your Code

```
1 s=input("Enter a string: ")
2 s=s.lower()
3 count={}
4 for ch in s:
5     if ch!=" ":
6         if ch in count:
7             count[ch]+=1
8         else:
9             count[ch]=1
10 for ch in sorted(count):
11     print(ch,"",count[ch])
```

Input

Hello world

Output

```
Enter a string: d 1
a 1
e 1
l 3
o 2
r 1
w 1
```

Submitted

SIMATS Online Programming Lab

PYTHON PROGRAMMING

Suba Shree K
192521396RD

Your Code

```
1 s=input()
2 vowels = "aeiou"
3 vowel_count=0
4 consonant_count=0
5
6 vowel_freq={}
7 consonant_freq={}
8 for ch in s.lower():
9     if ch.isalpha():
10        if ch in vowels:
11            vowel_count+=1
12            vowel_freq[ch]=vowel_freq.get(ch,0)+1
13        else:
14            consonant_count+=1
15            consonant_freq[ch]=consonant_freq.get(ch,0)+1
16 print("Vowels:",vowel_count)
17 print("Consonant:",consonant_count)
18 if vowel_freq:
19     max_vowel=max(vowel_freq,key=vowel_freq.get)
20     print("Most frequent vowel:",max_vowel,"(",vowel_freq[max_vowel],")")
21 else:
22     print("Most frequent vowel:None")
23 if consonant_freq:
24     min_consonant=min(consonant_freq,key=consonant_freq.get)
```

Input

Python Programming

Output

```
Vowels: 4
Consonant: 11
Most frequent vowel: i(2)
Least frequent consonant: (1)
Vowel to consonant Ratio: 4.00
```

SIMATS Online Programming Lab

PYTHON PROGRAMMING

Suba Shree K
192521396RD

Your Code

```
1 s=input()
2 if s=="":
3     print("Compressed:")
4     print("Decompressed:")
5 else:
6     c=""
7     count=1
8     for i in range(1,len(s)):
9         if s[i]==s[i-1]:
10            count+=1
11        else:
12            c+=s[i-1]+(str(count)if count>1 else "")
13            count=1
14 c+=s[-1]+(str(count)if count>1 else "")
15 if len(c)<len(s):
16     print("Compressed:",c)
17 d=""
18 i=0
19 while i<len(c):
20     ch=c[i]
21     i+=1
22     num=""
23     while i<len(c) and c[i].isdigit():
24         num+=c[i]
```

Input

aabbccdddee

Output

```
Compressed: a2b3c2d4e2
Decompressed: aabbccdddee
```

SIMATS Online Programming Lab

PYTHON PROGRAMMING

Suba Shree K
192521396RD

Top 5 Words: python(3), programming(2), language(2), data(2), code(1)

Flesch Reading Ease: 58.3 (Standard)

Your Code

```
1 import re
2 text=input()
3 words=re.findall(r'\b\w\b',text.lower())
4 sentences=len(re.findall(r'[.!?]',text))
5 paragraphs=1 if text.strip() else 0
6 print("Words:",len(words))
7 print("Sentences:",sentences)
8 print("Paragraphs:",paragraphs)
9 avg_word=sum(len(w) for w in words) / len(words) if len(words) else 0
10 avg_sent=len(words) / sentences if sentences else 0
11 print("Avg Word Length:", round(avg_word, 2))
12 print("Avg Sentence Length:", round(avg_sent, 2))
```

Input

Python is a powerful programming language.

Output

```
Words: 0
Sentences: 1
Paragraphs: 1
Avg Word Length: 0
Avg Sentence Length: 0.0
```